

Exc 01.

1/ We know that $1^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$$u_n = \frac{n(n+1)(2n+1)}{n^3}$$

$$u_n = \frac{(n+1)(2n+1)}{n^2}$$

$$u_{n+1} = \frac{(n+2)(2n+3)}{(n+1)^2}$$

$$u_{2n} = \frac{(2n+1)(4n^2+1)}{4n^2}$$

$$u_1 = 5 \text{ and } u_2 = \frac{15}{4} \text{ and } u_3 = \frac{28}{9}$$

$$2/ u_n = \frac{1}{(n-2)(n+1)}$$

$$u_{n+1} = \frac{1}{(n-1)(n+2)} \text{ and } u_{2n} = \frac{1}{(2n-2)(2n+1)}$$

$$u_3 = \frac{1}{4}$$

$$\text{and } u_5 = \frac{1}{18}$$

$$u_4 = \frac{1}{10}$$

$$3/ u_n = \frac{1 \times 3 \times 5 \times \dots \times (2n-1) \times (2 \times \dots \times 2n)}{2 \times 4 \times \dots \times (2n) \times (2 \times \dots \times 2n)}$$

$$= \frac{(2n)!}{(1 \times 2 \times \dots \times 2n)^2} \Rightarrow (1 \times 2 \times 3 \times \dots \times n \times 2 \times 2 \times \dots \times 2)$$

$$u_n = \frac{(2n)!}{2^{2n} \times (n!)^2}$$

$$u_{n+1} = \frac{(2n+2)!}{2^{2n+2} \times ((n+1)!)^2}$$

$$u_{2m} = \frac{(4m)!}{2^{4m} \times (2m!)^2}$$

$$u_1 = \frac{1}{2}$$

$$u_2 = \frac{3}{8}$$

$$u_3 = \frac{5}{16}$$

$$\begin{aligned} 4/ \quad u_{n+1} - u_n &= \frac{-1}{n+1} + \frac{1}{2n+1} + \frac{1}{2n+2} \\ &= \frac{1}{(2n+1)(n+1) \times 2} \end{aligned}$$

$$u_{n+1} = \frac{1}{2(2n+1)(n+1)} + u_n$$

$$u_{(2m-1)+1} = \frac{1}{2(4m-2+1)(2m)} + u_{2m-1}$$

$$= \frac{1}{2 \cdot 4m \cdot (4m-1)} + u_{2m-1}$$

$$u_1 = \frac{1}{1+1} = \frac{1}{2}$$

$$u_2 = \frac{1}{2(2 \times 1 + 1)(1+1)} + u_1 = \frac{1}{12} + \frac{1}{2} = \frac{7}{12}$$

$$u_3 = \frac{1}{2(2 \times 2 + 1)(2+1)} + u_2 = \frac{37}{60}$$

Ex 02;

$$\textcircled{1} \quad \frac{u_{n+1}}{u_n} = \frac{2}{n+1} \leq 1$$

(u_n) is Non increasing.

$\textcircled{2}$

$$u_{n+1} - u_n = 1 + 2(-1)^{n+1}$$

J.L:

$$n = 2k \mid k \in \mathbb{N}$$

$$u_{n+1} - u_n = 1 - 2 = -1 < 0$$

Ex: $n = 2k+1 / k \in \mathbb{N}_0$.

$$u_{n+1} - u_n = 3 > 0$$

(u_n) is not monotonic.

③

$$\begin{aligned} u_{n+1} - u_n &= \cancel{2n} + 2 + (-1)^{n+1} - \cancel{2n} \\ &\quad + (-1)^{n+1} \\ &= 2 + 2(-1)^{n+1} \end{aligned}$$

$$n = 2k / k \in \mathbb{N}$$

$$u_{n+1} - u_n = -2 + 2 = 0$$

$$n = 2k+1 / k \in \mathbb{N}_0$$

$$u_{n+1} - u_n = 2 + 2 = 4$$

$$\forall n \in \mathbb{N}$$

$$u_{n+1} - u_n \geq 0$$

(u_n) is non decreasing. $\square$

④

$$U_{n+1} - U_n = \cancel{n^2} + 1 + 2n - 2(-1)^{n+1} - \cancel{n^2} + 2(-1)^n$$

$$= 2n + 1 + 4(-1)^n$$

$$= 2n + 1 + 4(-1)^n$$

$$n = 2k \quad (k \in \mathbb{N})$$

$$U_{n+1} - U_n = 2(2k) + 1 + 4 > 0$$

$$n = 2k + 1 \quad (k \in \mathbb{N}_0)$$

$$U_{n+1} - U_n = 2n - 3$$

$$2n - 3 > 0$$

$$n > \frac{3}{2}$$

$$n \geq 2$$

$\forall n \geq 2$, (U_n) is increasing.

Ex 03:

$$a \in \mathbb{R} \\ |a| < 1$$

$$\lim_{n \rightarrow \infty} n a^n = 0$$

$\forall \varepsilon > 0, \exists N(\varepsilon)$ such that $\forall n > N(\varepsilon)$

$$|n a^n - 0| < \varepsilon$$

~~$$|n a^n| = n |a|^n < \varepsilon$$~~

We suppose that $|a| < 1$, and

$$\lim_{n \rightarrow \infty} n a^n = 0$$

$$\Leftrightarrow \exists \varepsilon > 0, \forall N(\varepsilon) \in \mathbb{N}, \exists n > N(\varepsilon)$$

~~$$\text{and } |n a^n - 0| < \varepsilon$$~~

$$n |a|^n > \varepsilon$$

$$|a|^n > \frac{\varepsilon}{n}$$

$$n \ln |a| > \ln \varepsilon - \ln n$$

$$\ln |a| > \frac{\ln \varepsilon}{n} - \frac{\ln n}{n}$$

$$\ln |a| > 0 \Rightarrow |a| > 1$$

contradiction

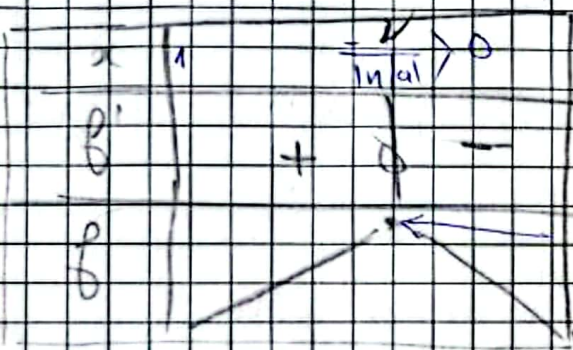
2nd method

let us consider $f(x) = x^2 |a|^x$

$x > 1$

$$f'(x) = x^2 e^{x \ln |a|} + 2x |a|^x x e^{x \ln |a|} - x e^{x \ln |a|} (2 + x \ln |a|)$$

$$1 + x \ln |a| = 0 \Rightarrow x = -\frac{1}{\ln |a|}$$



$$\forall x > 1, 0 < x^2 e^{x \ln |a|} < \frac{1}{e^2 \ln^2 |a|}$$

we have

$\Rightarrow \forall n \geq 1$, we have:

$$n e^{n \ln |a|} < \frac{1}{n}$$

$$n e^n < \frac{1}{n}$$

$\forall \varepsilon > 0, \exists N_0(\varepsilon) \in \mathbb{N} : \forall n > N_0(\varepsilon)$

$$\Rightarrow |n^n - a| = n |a|^n < \frac{1}{n} < \varepsilon$$

$$N_0(\varepsilon) = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1 \quad a > 1$$

$\forall \varepsilon > 0, \exists N_0(\varepsilon) \in \mathbb{N}, \forall n > N_0(\varepsilon)$

$$\Rightarrow |\sqrt[n]{a} - 1| < \varepsilon$$

$$|\sqrt[n]{a} - 1| = \sqrt[n]{a} - 1 < \varepsilon$$

$$\sqrt[n]{a} < 1 + \varepsilon$$

$$\frac{\ln a}{n} < \ln(1 + \varepsilon)$$

$$\Rightarrow n > \frac{\ln a}{\ln(1 + \varepsilon)}$$

$$N_0(\varepsilon) = \left\lceil \frac{\ln a}{\ln(1 + \varepsilon)} \right\rceil + 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

$$\sqrt[2]{2} = 1,4$$

$$\sqrt[3]{3} = 1,1$$

$$\sqrt[n]{n} = 1 + r_n$$

$$0 < r_n < 1$$

$$\sqrt[n]{n} = 1 + r_n$$

$$n = (1 + r_n)^n = \sum_{k=0}^n \binom{n}{k} r_n^k$$

$$= 1 + n r_n + \frac{n(n-1)}{2} r_n^2 + \dots$$

$$\sqrt[n]{n} = 1$$

For $n \in \mathbb{N}$, $\exists r_n > 0$ such that

$$\sqrt[n]{n} = 1 + r_n$$

$$\text{Then } n = (1 + r_n)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} r_n^k$$

$$= 1 + nr_n + \frac{n(n-1)}{2} r_n^2 + \dots + r_n^n$$

$$n \geq 1 + \frac{n \times (n-1)}{2} r_n^2$$

$$n-1 \geq \frac{n(n-1)}{2} (\sqrt[n]{n} - 1)^2$$

$$1 \geq \frac{n}{2} (\sqrt[n]{n} - 1)^2$$

$$(\sqrt[n]{n} - 1)^2 \leq \frac{2}{n}$$

$$|\sqrt[n]{n} - 1| \leq \frac{\sqrt{2}}{\sqrt{n}}$$

thus $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$ because

$$\forall \epsilon > 0, \exists N_0(\epsilon) = \left\lceil \frac{2}{\epsilon^2} \right\rceil + 1$$

$$\forall n \geq N_0(\epsilon) \Rightarrow$$

$$|\sqrt[n]{n} - 1| < \epsilon$$

$$\lim_{n \rightarrow \infty} \frac{\ln(n+1)}{\ln(n)} = 1$$

$\forall \epsilon > 0, \exists N_0(\epsilon) \in \mathbb{N}, \forall n \geq N_0(\epsilon)$

$$\left| \frac{\ln(n+1)}{\ln(n)} - 1 \right| < \epsilon$$

$$\frac{\ln(n+1)}{\ln(n)} - 1 = \frac{\ln(n(1 + \frac{1}{n}))}{\ln(n)} - 1$$

$$= \frac{\ln(n) + \ln(1 + \frac{1}{n})}{\ln(n)} - 1 = \frac{\ln(1 + \frac{1}{n})}{\ln(n)} < \frac{1/n}{\ln(n)} < \epsilon$$

$$\frac{\ln 2}{\ln n} < \epsilon \Rightarrow$$

$$\frac{1}{\ln n} < \frac{\epsilon}{\ln 2}$$

$$\frac{\ln 2}{\ln n} < \epsilon$$

$$n \geq e^{\frac{\ln 2}{\epsilon}}$$

$$N_0 = \left\lceil e^{\frac{\ln 2}{\epsilon}} \right\rceil + 1$$

$$\lim_{n \rightarrow \infty} \frac{\ln(1+e^n)}{2n} = \frac{1}{2}$$

$$\left| \frac{\ln(1+e^n)}{2n} - \frac{1}{2} \right| < \epsilon$$

$$\frac{\ln(1+e^n)}{2n} - \frac{1}{2} = \frac{\ln(e^n (\frac{1}{e^n} + 1))}{2n} - \frac{1}{2}$$

$$= \frac{n}{2n} + \frac{\ln(\frac{1}{e^n} + 1)}{2n}$$

$$= \frac{\ln(\frac{1}{e^n} + 1)}{2n} < \frac{\ln 2}{2n} < \epsilon$$

$$\frac{\ln 2}{2n} < \epsilon$$

$$\frac{1}{2n} < \frac{\epsilon}{\ln 2}$$

$$2n \geq \frac{\ln 2}{\epsilon}$$

$$n \geq \frac{\ln 2}{2\epsilon}$$

$$N_0(\epsilon) = \left\lceil \frac{\ln 2}{2\epsilon} \right\rceil + 1$$

$$\lim_{n \rightarrow \infty} \frac{2n+1}{3} = +\infty$$

$$\forall A > 0, \exists N_0(\varepsilon) \in \mathbb{N} \quad \forall n \geq N_0(\varepsilon)$$

$$\rightarrow \frac{2n+1}{3} > A$$

$$N_0(\varepsilon) = \left\lceil \frac{\ln A - \ln 3}{2 \ln 3} \right\rceil + 1$$

Ex 11:

Ex 4

$$\frac{1}{\sqrt{m}} \sum_{n=1}^m \left(\frac{\sin(n)}{n} \right) \leq \frac{1}{\sqrt{m}}$$

$$u_n = \frac{n^n}{n!}, \quad n \geq 0$$

ex0:04.

$$\ln u_n = \ln \left(\frac{n^n}{n!} \right) \geq 0$$

$$= n \ln(n) - \ln(n!).$$

$$= n \ln(n) - (\ln(n) + \ln(n-1) + \dots + \ln(2) + \ln(1))$$

$$= \underbrace{(\ln(n) + \ln(n) + \dots + \ln(n))}_{n \text{ times}} - (\ln(n) + \ln(n-1) + \ln(n-2) + \dots + \ln(2))$$

$$= (\ln(n) - \ln(n-1)) + (\ln(n) - \ln(n-2)) + \dots + (\ln(n) - \ln(2)) + \ln(n)$$

$$\lim_{n \rightarrow \infty} u_n \geq \lim_{n \rightarrow \infty} \ln(n) = +\infty$$

$$\text{f.e. } \lim_{n \rightarrow \infty} u_n = +\infty \Rightarrow u_n \rightarrow +\infty$$

21) $\lim_{n \rightarrow \infty} \frac{[n^2 \sin(\frac{1}{n})]}{2n}$

$$\text{we have } \sin \frac{1}{n} \approx \lim_{n \rightarrow \infty} \frac{1}{n}$$

$$\text{so we get } \frac{[n^2 \cdot \frac{1}{n}]}{2n+1} = \frac{[n]}{2n+1} \approx \lim_{n \rightarrow \infty} \frac{n}{2n} = \frac{1}{2}$$

$$U_n = \frac{1 + a + a^2 + \dots + a^n}{1 + b + b^2 + \dots + b^n} = \frac{\frac{1-a^{n+1}}{1-a}}{\frac{1-b^{n+1}}{1-b}} \rightarrow \frac{1-b}{1-a}$$

$$U_n = \frac{3^n + 7^n}{\left(\frac{3}{7}\right)^n + 1} = 1$$

$$u_n = \sum_{k=1}^n \frac{1}{\sqrt{k^2+1}}$$

$$\overline{p \Rightarrow q} \\ p \wedge \overline{q}$$

Def (Cauchy sequence):

(u_n) is of Cauchy $\stackrel{\text{def}}{=} \forall \varepsilon > 0, \exists n_0 \in \mathbb{N} / \forall n, m \in \mathbb{N} /$
 $((n > N_0 \wedge m > N_0) \Rightarrow |u_n - u_m| < \varepsilon)$

Negation: (u_n) is not of Cauchy $\stackrel{\text{def}}{=} \exists \varepsilon > 0, \forall N_0 \in \mathbb{N} / \exists n, m \in \mathbb{N} : n > N_0 \wedge m > N_0 \wedge |u_n - u_m| \geq \varepsilon$

$\exists \varepsilon > 0, \forall N_0 \in \mathbb{N} / \exists n, m \in \mathbb{N} : n > N_0 \wedge m > N_0 \wedge |u_n - u_m| \geq \varepsilon$

$$n = 2m$$

$$|u_n - u_m| = \left| \sum_{k=1}^{2m} \frac{1}{\sqrt{k^2+1}} - \sum_{k=1}^m \frac{1}{\sqrt{k^2+1}} \right| = \left| \sum_{k=m+1}^{2m} \frac{1}{\sqrt{k^2+1}} \right|$$

$$\sum_{k=m+1}^{12m} \frac{1}{\sqrt{k^2+1}} = \frac{1}{\sqrt{(m+1)^2+1}} + \frac{1}{\sqrt{(m+1)^2+1}} + \dots + \frac{1}{\sqrt{(9m)^2+1}}$$

$$= \frac{1}{\sqrt{(9m)^2+1}} + \frac{1}{\sqrt{(9m)^2+1}} + \dots + \frac{1}{\sqrt{(9m)^2+1}} = \frac{m}{\sqrt{4m^2+1}} \xrightarrow{m \rightarrow +\infty} \frac{1}{2} = \varepsilon.$$

Then, $(U_n)_n$ is not of Cauchy in $\mathbb{R}$. $\frac{1}{k^2} = \frac{1}{k \cdot k}$

Hence, $(U_n)_n$ diverge. $\left(\lim_{n \rightarrow +\infty} U_n = +\infty \right) \leq \frac{1}{k(k-1)}, k \geq 2.$

$$V_n = \sum_{k=0}^n \frac{k!}{(n+2)!}$$

For $k < n \Rightarrow k! < n!$

$$\Rightarrow \sum_{k=0}^n k! < \sum_{k=0}^n n! = (n+1)n!$$

So

$$V_n = \sum_{k=0}^n \frac{k!}{(n+2)!}$$

$$= \frac{1}{(n+2)!} \sum_{k=0}^n k!$$

$$< \frac{(n+1)n!}{(n+2)!}$$

$$= \frac{1}{n+2}$$

①

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} \geq 2 \sum_{k=1}^n \frac{1}{2\sqrt{k}}$$

$$= 2 \left(\frac{1}{2\sqrt{1}} + \frac{1}{2\sqrt{2}} + \dots + \frac{1}{2\sqrt{n}} \right) \geq$$

$$2 \left(\sqrt{2} - 1 + \sqrt{3} - \sqrt{2} + \sqrt{4} - \sqrt{3} + \dots + \sqrt{n} - \sqrt{n-1} \right)$$

$$= 2 \left(\sqrt{n+1} - 1 \right) \longrightarrow +\infty$$

$$\text{So: } \sum_{k=1}^n \frac{1}{\sqrt{k}} \longrightarrow +\infty$$

$$\tan\left(\frac{\theta}{2^n}\right) - 2^n \sin\left(\frac{\theta}{2^n}\right)$$

$$\left[1 - \frac{\sin\left(\frac{\theta}{2^n}\right)}{\tan\left(\frac{\theta}{2^n}\right)} \right]$$

$$\left(1 - \cos\left(\frac{\theta}{2^n}\right) \right)$$

$$\sin(2\theta) = 2$$

Exos:

a) $(u_{2n}), (u_{2n+1})$ convergent $\Leftrightarrow (u_n)$ CV to l .

$\Leftarrow$: If (u_n) converges to l .

(i.e. $\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N \Rightarrow |u_n - l| < \epsilon$).

As $2n+1 > 2n > n \geq N$ so.

$$|u_{2n} - l| < \epsilon \text{ and } |u_{2n+1} - l| < \epsilon.$$

So (u_{2n}) and (u_{2n+1}) convergent to l .

$\Rightarrow$: If (u_{2n}) and (u_{2n+1}) convergent to $l \Leftrightarrow$

$\forall \epsilon > 0, \exists N_1 \in \mathbb{N}, \forall n \geq N_1, |u_{2n} - l| < \epsilon$

$\forall \epsilon > 0, \exists N_2 \in \mathbb{N}, \forall n \geq N_2, |u_{2n+1} - l| < \epsilon$.

? $\exists N, \forall n \geq N, |u_n - l| < \epsilon$.

There exist $N = \max(N_1, N_2)$. $\forall n \geq N, |u_n - l| < \epsilon$.

For any $n \geq N$. we distinguish two cases:

• If n even ($n = 2p$) $2p \geq 2N \Leftrightarrow p \geq N$

$$|u_n - l| = |u_{2p} - l| < \epsilon$$

• If n odd ($n = 2p+1$) $2p+1 \geq 2N \Leftrightarrow p \geq N$

$$|u_n - l| = |u_{2p+1} - l| < \epsilon$$

So (u_n) converges to l .

b) $(u_n) \nearrow$ and (u_{2n}) CV, Then (u_n) CV.

$(u_n) \nearrow$ means That $\begin{cases} \lim_{n \rightarrow \infty} u_n = l \\ \text{or} \\ \lim_{n \rightarrow \infty} u_n = +\infty \end{cases}$ Since $u_n \nearrow$ Then $\lim_{n \rightarrow \infty} u_n = l$

$$\frac{1}{2\sqrt{k+1}} \leq \sqrt{k+1} - \sqrt{k} = \frac{1}{\sqrt{k+1} + \sqrt{k}} \leq \frac{1}{2\sqrt{k}}$$

$$2(\sqrt{k+1} - \sqrt{k}) \leq \frac{1}{\sqrt{k}}$$

$$2 \sum (\sqrt{k+1} - \sqrt{k}) \leq \sum \frac{1}{\sqrt{k}}$$

$$2(\sqrt{n+1} - 1) \leq \sum \frac{1}{\sqrt{k}}$$

$$\lim \sum \frac{1}{\sqrt{k}} \rightarrow \infty$$

$$U_n = \sum \frac{1}{k^2} < 1 + \sum \frac{1}{k(k-1)} < 1 + \sum_{k=2}^{\infty} \frac{1}{k-1} - \frac{1}{k}$$

EXERC.

$$a_n = n \left(\frac{1}{n} \right)^n$$

$$n = 2k$$

$$a_n = a_{2k} = (2k)^{2k} \xrightarrow{k \rightarrow \infty} \infty$$

$$n = 2k+1$$

$$a_n = a_{2k+1} = (2k+1)^{2k+1} \xrightarrow{k \rightarrow \infty} \infty$$

$$\lim a_n = +\infty$$

$$\lim a_n = 0$$

$$a_n = 1 + n \sin \frac{n\pi}{2}$$

$$\text{for } n = 4k$$

$$a_{4k} = 1 + 4k \sin(2k\pi) \rightarrow 1$$

$$a_{4k+1} = 1 + (4k+1) \sin\left(\frac{4k\pi + \pi}{2}\right) \rightarrow +\infty$$

$$a_{4k+2} = 1 + (4k+2) \sin\left(\frac{4k\pi + 2\pi}{2}\right) \rightarrow 1$$

$$a_{4k+3} = 1 + (4k+3) \sin\left(\frac{4k\pi + 3\pi}{2}\right) \rightarrow -\infty$$

$$\lim a_n = +\infty$$

$$\lim a_n = -\infty$$

$$a_n = \left(2 \cos \frac{2n\pi}{3} \right)^n$$

$$a_{3k} = \left(2 \cos(2k\pi) \right)^{3k} = 2^{3k} \rightarrow +\infty$$

$$a_{3k+1} = \left(2 \cos\left(2k\pi + \frac{2\pi}{3}\right) \right)^{3k+1} = (-1)^{3k+1} \rightarrow -1$$

$$a_{3k+2} = \left(2 \cos\left(2k\pi + \frac{4\pi}{3}\right) \right)^{3k+2} = (-1)^{3k+2} \rightarrow 1$$

$$\lim a_n = +\infty$$

$$\lim a_n = -1$$

Ex 07:

U_n is CV $\Rightarrow U_n$ is Cauchy

U_n Cauchy $\Leftrightarrow$

$$U_n \rightarrow l$$

$$U_{n+1} \rightarrow l$$

Using Algebraic rule

$$U_{n+1} - U_n \rightarrow l - l = 0$$

Ex 07:

$$U_n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}$$

$$0 \leq U_n = \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \cdots \frac{2n-1}{2n} \leq 1$$

so U_n is bounded

$$\frac{U_{n+1}}{U_n} = \frac{2n+1}{2n+2} < 1$$

$\Rightarrow U_n$ decreasing and lower bounded

so U_n CV.

$$S_n = \sum \frac{1}{k^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots$$

$$< 1, 2 + \sum_{k=1}^{\infty} \frac{1}{(k+1)(k+2)}$$

$$< 1, 2 + \sum_{k=1}^{\infty} \frac{1}{k+2} - \frac{1}{k+1}$$

$$< 1, 2 + 1 - \frac{1}{n-1} < 2, 2$$

$$\forall n \in \mathbb{N}$$

$$0 \leq S_n \leq 2, 2$$

$$S_{n+1} - S_n = \frac{1}{(n+1)^2} > 0$$

Ex 08:

$$\begin{cases} U_{n+1} = \sqrt{2+U_n} \\ U_1 = \sqrt{2} \end{cases}$$

$$f(x) = \sqrt{2+x}$$

$$f'(x) = \frac{1}{2\sqrt{2+x}} > 0 \quad \forall x > 0$$

$$\forall n \in \mathbb{N} : U_n > 0$$

$$U_1 = \sqrt{2} > 0$$

$$U_n > 0 \Rightarrow U_{n+1} > 0$$

$$\sqrt{U_{n+1}} > \sqrt{2} \Rightarrow U_{n+1} > 0$$

one has

$$U_n - U_{n+1} = \sqrt{2+U_n} - \sqrt{2} > 0$$

U_n is increasing

by ind

$$U_1 = \sqrt{2} < 2 \text{ true}$$

$$U_n \leq 2 \text{ and}$$

$$U_{n+1} = \sqrt{2+U_n} \leq \sqrt{2+2} = 2$$

U_n and upper bounded, so CV

EX 10.10

$$\begin{cases} U_0 = \alpha \\ U_{n+1} = \frac{U_n}{2} + 1 \end{cases}$$

the associate function.

$$f(x) = \frac{1}{2}x + 1 \quad f'(x) = \frac{1}{2} > 0$$

thus the sequence U_n is
monotonic of the same variation
as $U_1 - U_0$

$$U_1 - U_0 = \frac{1}{2}U_0 + 1 - U_0 = 1 - \frac{1}{2}U_0 = 1 - \frac{\alpha}{2}$$

let

$$\alpha \leq 2$$

$$U_1 - U_0 = 1 - \frac{\alpha}{2} \geq 0$$

U_n is nondecreasing ($\forall n, U_{n+1} \geq U_n$)

let us show that $U_n \leq 2$

$$U_0 = \alpha \leq 2$$

suppose that $U_n \leq 2$

$$\frac{U_n}{2} + 1 \leq 2 \quad \text{so}$$

$$U_{n+1} \leq 2$$

then U_n is nondecreasing and upper
bounded by 2 means convergent

$$l = f(l) \Leftrightarrow l = \frac{l}{2} + 1$$

$$\Leftrightarrow l = 2$$

$$x > 2$$

U_n is decreasing

let us show that $U_n > 2$

$U_0 = x > 2$ suppose that $U_n > 2$

$$\Rightarrow U_{n+1} = \frac{U_n}{2} + 1 > 2$$

decreasing and lower bounded mean

convergent then

$$\lim_{n \rightarrow \infty} U_n = l = 2$$

$$\begin{cases} U_0 = 1 \\ U_{n+1} = \sqrt{U_n + 2} \end{cases}$$

$$U_1 = \sqrt{1+2} = \sqrt{3}; \quad U_2 = \sqrt{\sqrt{3}+2} \dots$$

Suppose $U_n \xrightarrow{n \rightarrow \infty} l$ so, $l = \sqrt{l+2}$

$$l^2 - l - 2 = 0 \Rightarrow \begin{cases} l = 2 \\ l = -1 \text{ refused.} \end{cases}$$

$$U_n > 0 \text{ so } \boxed{l = 2}$$

$$U_0 = 1; \quad U_1 = \sqrt{3} = 1.73; \quad U_2 = \sqrt{\sqrt{3}+2} = 1.93$$

so $\{U_n\}$ is monotonically increasing

We have $U_0 = 1$ and $U_n \rightarrow 2$.

so U_n of course should be monotonically increasing

1.1 1.1 1.1

1.1 1.1 1.1

1.1 1.1 1.1

Exercise 11:

we've $u_n > 0$ since $\theta \in]0, \pi/2[$. Using $\sin 2\theta = 2 \cos \theta \sin \theta$!

$$\text{Then, } \frac{u_{n+1}}{u_n} = \frac{2^{n+1} \sin \frac{\theta}{2^{n+1}}}{2^n \sin \frac{\theta}{2^n}} = 2 \frac{\sin \frac{\theta}{2^{n+1}}}{2 \sin \frac{\theta}{2^{n+1}} \cos \frac{\theta}{2^{n+1}}}$$

$$= \frac{1}{\cos \frac{\theta}{2^{n+1}}} > 1.$$

Then, (u_n) is ~~de~~ increasing.

Also,

$$\frac{v_{n+1}}{v_n} = \frac{2^{n+1} \frac{\sin \frac{\theta}{2^{n+1}}}{\cos \frac{\theta}{2^{n+1}}}}{2^n \frac{\sin \frac{\theta}{2^n}}{\cos \frac{\theta}{2^n}}} = 2 \frac{\sin \frac{\theta}{2^{n+1}}}{\cos \frac{\theta}{2^{n+1}}} \cdot \frac{\cos \frac{\theta}{2^n}}{2 \cdot \cos \frac{\theta}{2^{n+1}} \sin \frac{\theta}{2^{n+1}}}$$

$$= \frac{\cos \frac{\theta}{2^n}}{\cos^2 \frac{\theta}{2^{n+1}}} = \frac{\cos^2 \frac{\theta}{2^{n+1}} - \sin^2 \frac{\theta}{2^{n+1}}}{\cos^2 \frac{\theta}{2^{n+1}}} = 1 - \tan^2 \frac{\theta}{2^{n+1}} < 1.$$

$$(\cos 2\theta = \cos^2 \theta - \sin^2 \theta)$$

Then, (v_n) is decreasing.

furthermore,

$$\lim_{n \rightarrow +\infty} (u_n - v_n) = \lim_{n \rightarrow +\infty} 2^n (\sin \frac{\theta}{2^n} - \tan \frac{\theta}{2^n})$$

$$= \lim_{n \rightarrow +\infty} 2^n \tan \frac{\theta}{2^n} (\cos \frac{\theta}{2^n} - 1)$$

$$= \lim_{n \rightarrow +\infty} -2^{n+1} \tan \frac{\theta}{2^n} \sin^2 \frac{\theta}{2^{n+1}}$$

$$\lim_{n \rightarrow +\infty} -2 \frac{\theta}{2^n} \cdot \frac{\theta^2}{2^{2n+1}} \rightarrow 0$$

(1)

$$\text{Then, } \lim_{n \rightarrow \infty} (u_n - y_n) = 0$$

Conclusion: $(u_n)_n$ and $(y_n)_n$ are adjacent.

2. Second example: Same way.

Exercise 12:

$$\begin{aligned} \text{a) } |u_{n+1} - u_n| &\leq k |u_n - u_{n-1}| \leq k^2 |u_{n-1} - u_{n-2}| \leq \dots \\ &\leq \dots \leq k^n |u_1 - u_0| \quad (\text{by induction}). \end{aligned}$$

b) For all $n, m \in \mathbb{N}$, $n > m$, we get:

$$|u_n - u_m| = |u_n - u_{n-1} + u_{n-1} - u_{n-2} + \dots + u_{m+1} - u_m| \leq$$

$$\leq |u_n - u_{n-1}| + |u_{n-1} - u_{n-2}| + \dots + |u_{m+1} - u_m| \leq$$

$$\leq k^{n-1} \sum_{p=m+1}^n k^p |u_1 - u_0| \leq k^{m+1} \frac{1 - k^{n-m}}{1 - k} |u_1 - u_0| \leq$$

$$\leq k^{m+1} |u_1 - u_0| \xrightarrow{m \rightarrow \infty} 0 \quad \text{since } 0 < k < 1.$$

Then, $(u_n)_n$ is of Cauchy in $\mathbb{R}$. $(u_n)_n$ is
 a.s.t.e.

Exercise 13: a) $U_n = \sum_{k=1}^n \frac{1}{k^2}$.

let $n, m \in \mathbb{N}$, $n > m$:

$$|U_n - U_m| = \left| \sum_{k=m+1}^n \frac{1}{k^2} \right| \leq \sum_{k=m+1}^n \frac{1}{k^2} =$$

$$\frac{1}{(m+1)^2} + \frac{1}{(m+2)^2} + \dots + \frac{1}{n^2} <$$

$$< \frac{1}{m(m+1)} + \frac{1}{(m+1)(m+2)} + \dots + \frac{1}{n(n-1)} < \left(\frac{1}{k^2} < \frac{1}{k-1} - \frac{1}{k} \right)$$

$$< \frac{1}{m} - \frac{1}{m+1} + \frac{1}{m+1} - \frac{1}{m+2} + \dots + \frac{1}{n-1} - \frac{1}{n} <$$

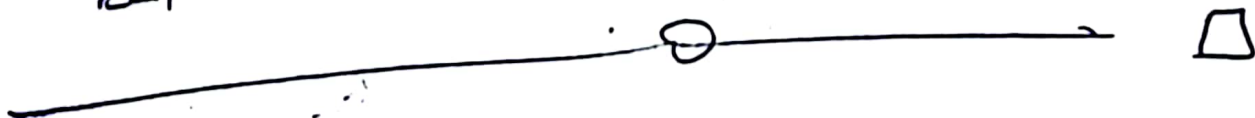
$$< \frac{1}{m} - \frac{1}{n} < \frac{1}{m} \xrightarrow{m \rightarrow +\infty} 0.$$

then $(U_n)_n$ is of Cauchy in $\mathbb{R}$. It is

then write.

b) $\sum_{k=1}^n \frac{1}{3^k}$ is also write by the same way.

c) $\sum_{k=1}^n \frac{1}{\sqrt{k}}$ is divergent, as in Exercise 4.



$$u_n \rightarrow l \Rightarrow \ln(u_n) \xrightarrow{n \rightarrow \infty} \ln(l).$$

Hence $\ln v_n \rightarrow \ln(l)$ by exercise 15.

$$\text{Thus, } v_n \xrightarrow{n \rightarrow \infty} l. \quad \square$$

(c) with a same way. $\square$

Exercise 14: $u_n = \sum_{k=1}^n \frac{(-1)^k}{k}$

$$u_{n+1} = \sum_{k=1}^{n+1} \frac{(-1)^k}{k}$$

we've: $u_{n+2} - u_n = \frac{-1}{n+1} + \frac{1}{n+2} = \frac{-1}{(n+1)(n+2)} < 0$

and $u_{n+3} - u_{n+1} = \frac{-1}{n+3} + \frac{1}{n+2} = \frac{1}{(n+3)(n+2)} > 0$

Hence, (u_n) is decreasing and (u_{n+1}) is increasing.

Further, we've:

$$\lim_{n \rightarrow \infty} (u_{n+1} - u_n) = \lim_{n \rightarrow \infty} \frac{-1}{n+1} = 0$$

Hence, (u_n) and (u_{n+1}) are convergent to the same limit l . Then, we get that $u_n \rightarrow l, n \rightarrow \infty.$

(5)

$\square$

Exercise 15: $U_n \xrightarrow{n \rightarrow \infty} l \Rightarrow U_n \xrightarrow{n \rightarrow \infty} l$. ??

$\lim_{n \rightarrow \infty} U_n = l \Rightarrow \forall \varepsilon > 0, \exists N_0 \in \mathbb{N} / \forall n \geq N_0: |U_n - l| < \varepsilon/2$... (*)

$$|U_n - l| = \left| \sum_{k=1}^n \frac{U_k}{n} - l \right| = \frac{1}{n} \left| \sum_{k=1}^n U_k - n l \right|$$

$$= \frac{1}{n} |U_1 + U_2 + \dots + U_{N_0} + U_{N_0+1} + U_{N_0+2} + \dots + U_n - n l|$$

$$= \frac{1}{n} |(U_1 - l) + (U_2 - l) + \dots + (U_{N_0} - l) + (U_{N_0+1} - l) + \dots + (U_n - l)|$$

$$\leq \frac{1}{n} \sum_{k=1}^{N_0} |U_k - l| + \frac{1}{n} \sum_{k=N_0+1}^n |U_k - l|$$

we use: $\sum_{k=1}^{N_0} |U_k - l| = \lambda, \lambda \in \mathbb{R}$.

Then $\frac{1}{n} \sum_{k=1}^{N_0} |U_k - l| = \frac{\lambda}{n} \xrightarrow{n \rightarrow \infty} 0$

Hence, $\forall \varepsilon > 0, \exists N_1 \in \mathbb{N} / \forall n \geq N_1: \frac{\lambda}{n} < \varepsilon/2$... (1)

let then $N = \max(N_0, N_1)$. Thus,

$\forall n \geq N, \frac{1}{n} \sum_{k=1}^n |U_k - l| < \varepsilon/2$ and by (*):

$$\frac{1}{n} \sum_{k=N_0+1}^n |U_k - l| \leq \frac{1}{n} (n - N) \varepsilon/2 \leq \frac{n}{n} \frac{\varepsilon}{2} = \frac{\varepsilon}{2}$$

... (2)

(3)

By (1) and (2), we get:

$$\forall n, N: |V_n - l| < \frac{\epsilon}{2} + \frac{1}{n}(n-N)\frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

finally, $\lim_{n \rightarrow \infty} V_n = l \quad \dots \square$.

Exercise 16: a) Direct consequence of Exercise 15.

Put $W_n = U_n - U_{n-1}, n \geq 1$. Then, $W_n \xrightarrow[n \rightarrow \infty]{} l$.

$$V_n = \frac{W_1 + W_2 + \dots + W_n}{n} \xrightarrow[n \rightarrow \infty]{} l \text{ by ex 15,}$$

$$V_n = \frac{U_1 - U_0 + U_2 - U_1 + \dots + U_{n-1} - U_{n-2} + U_n - U_{n-1}}{n}$$

$$V_n = \frac{U_n - U_0}{n} \xrightarrow[n \rightarrow \infty]{} l.$$

$$\Rightarrow \frac{U_n}{n} - \frac{U_0}{n} \xrightarrow[n \rightarrow +\infty]{} l$$

$$\Rightarrow \frac{U_n}{n} \xrightarrow[n \rightarrow +\infty]{} l \quad \square$$

b) Also, a direct consequence of Exercise 15.

$$U_n \rightarrow l \Rightarrow \sqrt[n]{U_1 U_2 \dots U_n} = l$$

Put $W_n = \sqrt[n]{U_1 U_2 \dots U_n}$. Then,

$$V_n = \ln W_n = \frac{\ln U_1 + \ln U_2 + \dots + \ln U_n}{n}$$

(4)

Exercise 17:

1. $l < 1$. Let $k \in \mathbb{R}$ fixed such that $l < k < 1$.

Since $\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = l$, there exists $N_0 \in \mathbb{N}$:

$$\forall n \geq N_0: \sqrt[n]{u_n} < k \quad \left(k = l + \varepsilon \text{ for } \begin{array}{l} l + \varepsilon < 1 \\ \Rightarrow 0 < \varepsilon < 1 - l \end{array} \right)$$

$$\Rightarrow \forall n \geq N_0: 0 \leq u_n < k^n \rightarrow 0 < k < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = 0$$

2. $l > 1$: $\exists N_0 \in \mathbb{N} / \forall n \geq N_0: \sqrt[n]{u_n} > l - \varepsilon > 1$ for some $\varepsilon > 0$

$$\Rightarrow \forall n \geq N_0: \sqrt[n]{u_n} > (l - \varepsilon) > 1 \quad \text{S.t.: } 0 < \varepsilon < l - 1 \quad (l > 1)$$

$$\Rightarrow \forall n \geq N_0: u_n > (l - \varepsilon)^n \rightarrow +\infty \quad l - \varepsilon > 1.$$

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = +\infty$$

3. $l = 1$: not conclusive. Counter-example:

$$u_n = n \text{ and } v_n = 1/n, n \in \mathbb{N}.$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \sqrt[n]{v_n} = 1 = l.$$

$$\text{However, } \lim_{n \rightarrow \infty} u_n = +\infty \text{ and } \lim_{n \rightarrow \infty} v_n = 0$$

Exercise 18:

$$a) \lim_{n \rightarrow +\infty} \frac{u_{n+1}}{u_n} = l \Rightarrow \lim_{n \rightarrow +\infty} \sqrt[n]{u_n} = l$$

It is a consequence of exercise 16, (a).

Put: ~~$V_n = \ln\left(\frac{u_{n+1}}{u_n}\right) = \ln(u_{n+1}) - \ln(u_n)$~~

Put $V_n = \ln(u_n)$. Then:

$$V_{n+1} - V_n = \ln(u_{n+1}) - \ln(u_n) = \ln\left(\frac{u_{n+1}}{u_n}\right)$$

$$\Rightarrow \lim_{n \rightarrow +\infty} (V_{n+1} - V_n) = \ln(l) \text{ by assumption.}$$

By exercise 16, (a), $\lim_{n \rightarrow +\infty} \frac{V_n}{n} = \ln l$ also.

That is, $\lim_{n \rightarrow +\infty} \frac{\ln(u_n)}{n} = \ln l$

$$\Rightarrow \lim_{n \rightarrow +\infty} \frac{1}{n} \ln(u_n) = \lim_{n \rightarrow +\infty} \ln(u_n)^{1/n} = \ln l$$

$$\Rightarrow \lim_{n \rightarrow +\infty} (u_n)^{1/n} = l. \quad \square$$

b). $\lim_{n \rightarrow +\infty} \frac{n}{\sqrt[n]{n!}}$. Put $u_n = \frac{n}{\sqrt[n]{n!}}$.

Then $u_n = \sqrt[n]{V_n}$ where $V_n = \frac{n}{n!}$.

we've then:

$$\begin{aligned}\frac{V_{n+1}}{V_n} &= \frac{(n+1)^{n+1}}{n^n} \cdot \frac{n!}{(n+1)!} = \frac{(n+1)^{n+1}}{n^n} \cdot \frac{1}{n+1} \\ &= \frac{(n+1)^n}{n^n} = \left(1 + \frac{1}{n}\right)^n \xrightarrow{n \rightarrow +\infty} e\end{aligned}$$

$$\Rightarrow \lim_{n \rightarrow +\infty} \sqrt[n]{V_n} = \lim_{n \rightarrow +\infty} U_n = e.$$

$$U_n = \frac{1}{n} \sqrt[n]{n(n+1) \dots (2n)}.$$

$$U_n^n = \frac{n(n+1) \dots (2n)}{n^n} = X_n$$

$$\begin{aligned}\frac{X_{n+1}}{X_n} &= \frac{(n+1)(n+2) \dots (2n+2)}{(n+1)^{n+1}} \cdot \frac{n^n}{n(n+1) \dots (2n)} \\ &= \frac{(2n+1)(2n+2)}{n(n+1)} \cdot \frac{n^n}{(n+1)^n} \\ &= \frac{(2n+1)(2n+2)}{n(n+1)} \cdot \frac{1}{\left(1 + \frac{1}{n}\right)^n} \rightarrow \frac{4}{e}\end{aligned}$$

$$\lim_{n \rightarrow +\infty} \frac{X_{n+1}}{X_n} = \frac{4}{e} \Rightarrow \lim_{n \rightarrow +\infty} U_n = \frac{4}{e}$$

(5)

$$a_n = \frac{a^n}{n!} \quad (a > 0)$$

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow +\infty} \frac{a^{n+1}}{a^n} \cdot \frac{n!}{(n+1)!} = \lim_{n \rightarrow +\infty} \frac{a}{n+1} = 0 < 1$$

Then since $a_n \geq 0, n \in \mathbb{N}$, then $\lim_{n \rightarrow +\infty} a_n = 0$

$$V_n = n^q \cos^n \theta, \quad 0 < \theta < \pi/2.$$

$$\frac{V_{n+1}}{V_n} = \frac{(n+1)^q \cos^{n+1}(\theta)}{n^q \cos^n \theta} = \left(\frac{n+1}{n}\right)^q \cos \theta \rightarrow 1^q \cos \theta = \cos \theta < 1$$

Since $0 < \theta < \pi/2$

Then, $\lim_{n \rightarrow +\infty} V_n = 0 \quad (V_n \geq 0)$.

$$w_n = \frac{2^{n^2}}{(n!)^n}$$

$$\sqrt[n]{w_n} = \frac{2^n}{n!} \rightarrow 0 < 1 \quad \left(2^n \leq n!, n \in \mathbb{N}, \text{ by induction} \right)$$

$$\Rightarrow \lim_{n \rightarrow +\infty} w_n = \infty$$

ملاحظة : الصفحة مخصصة لحل التمرين 19 ، الاس

ئلة الاولى هي نفسها الخاصة بالتمرين 17

$$x_n = \sqrt{2n} - \sqrt{2n-1}$$

$$y_n = \sqrt{n+(-1)^{n+1}} - \sqrt{n}$$

and

$$\Delta = \{x_n : n \in \mathbb{N}\}.$$

$$\cup = \{y_n : n \in \mathbb{N}\}.$$

(a) Know that $x_n > 0 \forall n$

$$x_1 = \sqrt{2} - 1 = 0.41$$

$$\Rightarrow 0 < x_n \leq x_1$$

Δ is bounded.

$$\sup \Delta = x_1 = 0.41$$

no $\max \Delta$

$$\inf \Delta = \min \Delta = 0.$$

(b)

n is odd

$$y_n = \sqrt{n+1} - \sqrt{n}$$

n is even

$$y_n = \sqrt{n-1} - \sqrt{n}$$

$$y_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} \sim \frac{1}{2\sqrt{n}} > 0$$

$$y_n = \frac{-1}{\sqrt{n+1} + \sqrt{n}} \sim \frac{-1}{2\sqrt{n}} < 0$$

So max for $y_n = \sqrt{n+1} - \sqrt{n}$

and min for $y_n = \sqrt{n-1} - \sqrt{n}$.

$$y_1 = \sqrt{2} - 1$$

$$\Rightarrow y_n < y_1$$

$$\max U = \sqrt{2} - 1 = \sup U.$$

and

$$\min U \text{ from } y_n = \sqrt{n-1} - \sqrt{n}$$

$$y_2 = 1 - \sqrt{2}$$

$$\text{So } \min U = \inf U = 1 - \sqrt{2}$$

Exo 21, $U_0 = 0$ then

$$U_n = \frac{1}{2} \sin(U_{n-1}) + \frac{1}{3} \cos(U_{n-1})$$

- prove that (U_n) is bounded.

Know that

$$\begin{cases} -1 \leq \sin(U_{n-1}) \leq 1 \\ \text{and} \\ -1 \leq \cos(U_{n-1}) \leq 1 \end{cases}$$

$$\Rightarrow \begin{cases} -\frac{1}{2} \leq \frac{1}{2} \sin(U_{n-1}) \leq \frac{1}{2} \\ \text{and} \\ -\frac{1}{3} \leq \frac{1}{3} \cos(U_{n-1}) \leq \frac{1}{3} \end{cases}$$

$$\Rightarrow -\frac{5}{6} < U_n < \frac{5}{6}$$

so U_n is bounded.

- prove that U_n is a seq of Cauchy

$$|U_n - U_{n-1}| = \left| \frac{1}{2} \sin(U_{n-1}) + \frac{1}{3} \cos(U_{n-1}) - U_{n-1} \right|$$

$$\simeq \left| \frac{1}{2} U_{n-1} + \frac{1}{3} - U_{n-1} \right|$$

$$= \left| -\frac{1}{2} U_{n-1} + \frac{1}{3} \right| \simeq \frac{1}{3}$$

← from small ϵ

$$|U_n - U_{n-1}| \simeq \frac{1}{3} = \epsilon.$$

so U_n is a Cauchy sequence.
this means that U_n is
convergent to l .

$$\begin{aligned} \sin x &\simeq x \\ \cos x &\simeq 1 \end{aligned}$$